Employee Performance Analytics Dashboard

Workforce Profile, Performance, Pay and Promotion – Power BI Project Report

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Project Title	Employee Performance Analytics Dashboard (Employees_Performance_Analysis_safe)
Tool Used	Microsoft Power BI Desktop (Power Query, DAX, Report View)

1. Dataset Description

The project uses a single employee dataset, loaded into Power BI as the table **employee_performance**. It holds records of **5,000 employees** across four departments, covering their job level, work location, tenure, performance rating, salary, absence and promotion status.

Field	Description
Employee ID	Unique employee identifier (for example E1019)
Department	Engineering, HR, Marketing or Sales
Gender	Female, Male or Non-binary
Job Level	Seniority level of the employee: 1, 2 or 3
Work Location	In-office, Hybrid or Remote
Tenure Bucket	Years of service grouped as 0-2, 3-5, 6-10, 11-15 and 16+ years
Performance Score	Rating from 1 (lowest) to 5 (highest)
Salary	Employee pay
Absence Days	Number of days the employee was absent
Promotion	Whether the employee has been promoted (used for the promotion rate)

2. Objectives

- Build a four-page interactive HR dashboard in Power BI covering Overview, Performance, Pay & Promotion and Employee Drill-down.
- Profile the workforce by department, gender, work location, job level and tenure.
- Measure average performance and absence, and understand how they differ across departments, locations and job levels.
- Analyse salary levels and the promotion rate across job level, performance, gender, tenure and department.
- Give managers an employee-level drill-down so any individual record can be compared with the group.

3. Data Cleaning Performed

The following preparation steps were carried out in Power Query before building the report:

- Imported the employee dataset and promoted the first row to column headers.
- Set correct data types (text for IDs and categories, whole numbers for Job Level and Performance Score, decimals for Salary and Absence Days).
- Checked for null, blank and duplicate Employee ID records and removed or corrected them.
- Standardised category values (department, gender, work location) and trimmed extra spaces.
- Created the helper column Tenure Bucket to group years of service into five logical bands.
- Sorted the Tenure Bucket and Performance Score columns in a logical order for the charts.

4. Data Model

The model is a flat, single-table design: employee_performance is the only table, so no relationships are needed and every visual on every page reads from it. The report has four pages, switched through the page tabs at the bottom. Four slicers at the top of every page (Department, Work Location, Gender or Job Level, and Tenure Bucket) filter all visuals together, and *Keep all filters* is turned on.

Page	Main visuals
Overview	5 KPI cards, employees by department (bar), gender split (donut), work location, performance score, job level and tenure (columns)
Performance	4 KPI cards, avg performance by department and by work location, avg absence days by performance score and by department, performance mix by job level
Pay & Promotion	4 KPI cards, avg salary by job level / performance score / gender, promotion rate by tenure (line) and by department (bar), Department Scorecard table
Employee Drill-down	4 KPI cards, Employee ID wise performance table, avg performance by job level, tenure bucket and work location

5. DAX Measures

The KPI cards and chart values are driven by simple DAX measures such as the following:

```
Total Employees = DISTINCTCOUNT ( employee_performance[Employee_ID] )
Avg Performance = AVERAGE ( employee_performance[Performance_Score] )
Avg Salary      = AVERAGE ( employee_performance[Salary] )
Avg Absence Days = AVERAGE ( employee_performance[Absence_Days] )
Promoted Employees = CALCULATE ( [Total Employees],
                        employee_performance[Promotion] = 1 )
Promotion Rate  = DIVIDE ( [Promoted Employees], [Total Employees] )
```

The promotion rate is the number of promoted employees divided by the total number of employees (about 20.2% for the whole company). The same measures, filtered by department, job level, tenure or gender, produce every chart on the dashboard. *Column names in the formulas are representative; they follow the fields listed in Section 1.*

6. Dashboard Screenshots

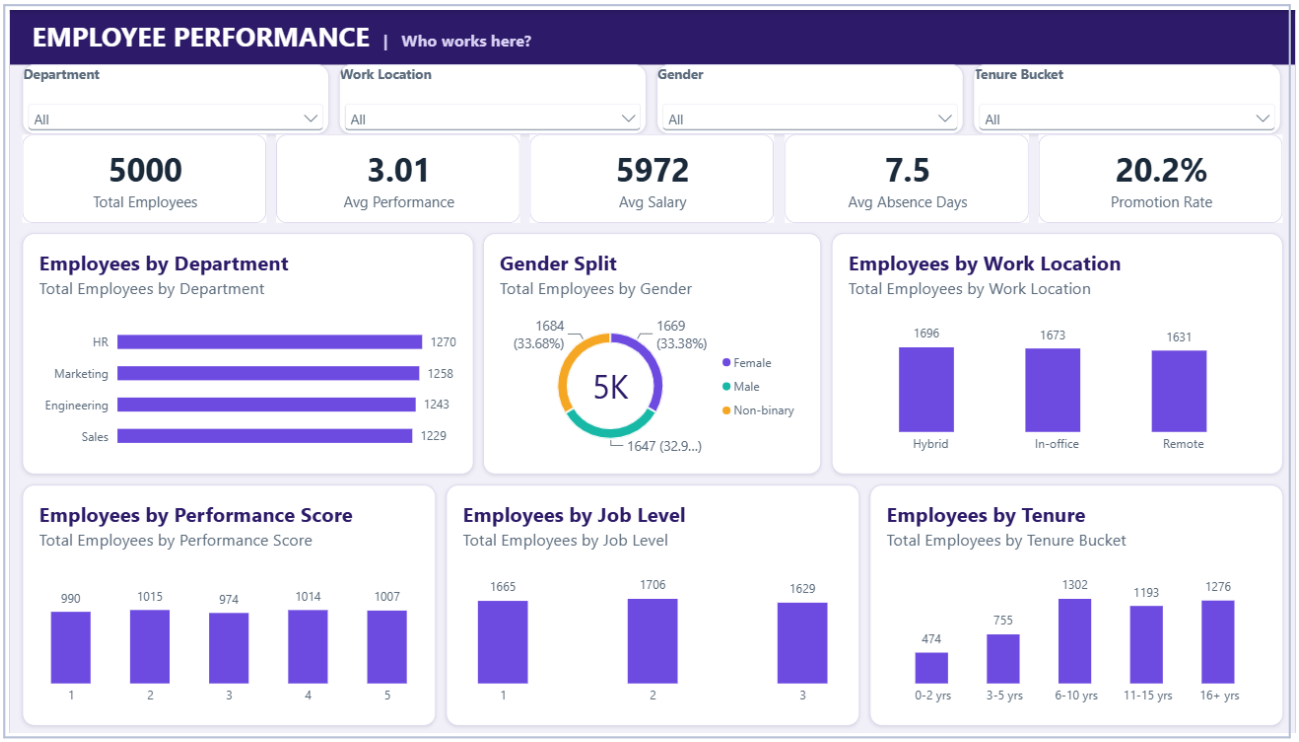


Figure 1: Overview page – Who works here?

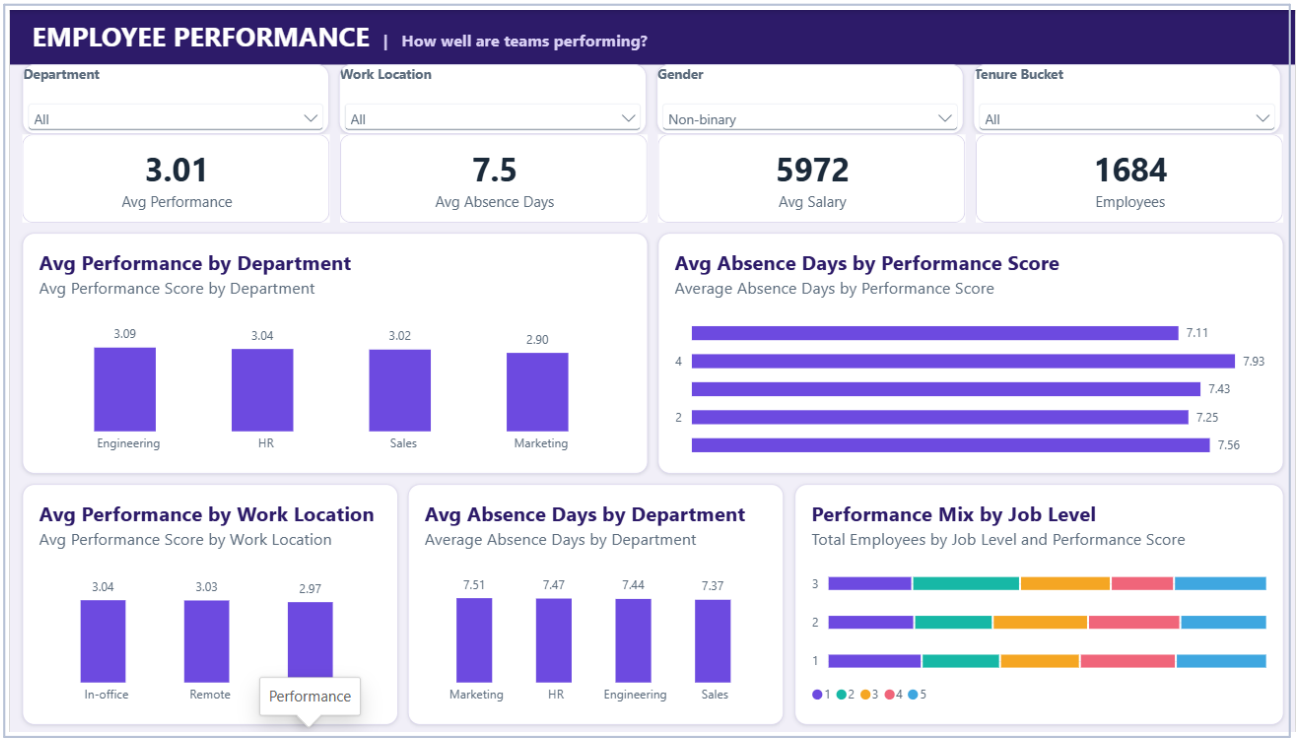


Figure 2: Performance page – How well are teams performing?

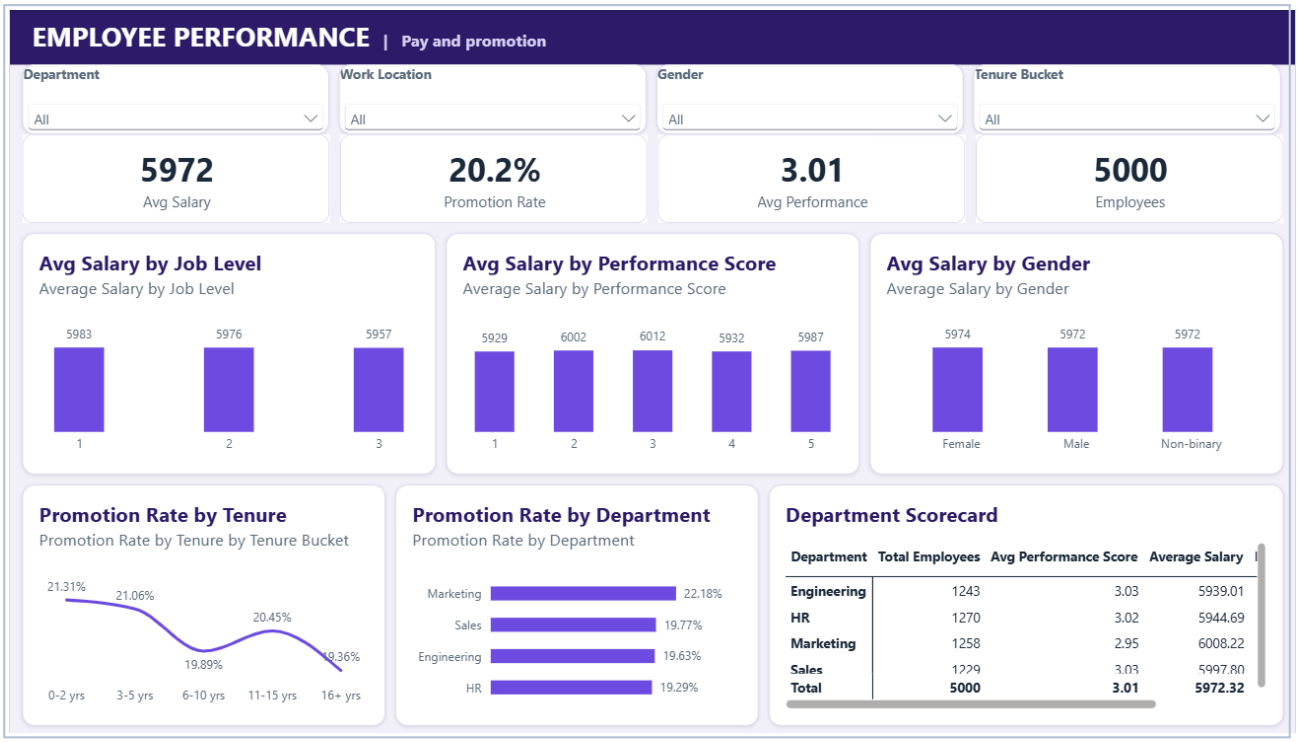


Figure 3: Pay & Promotion page

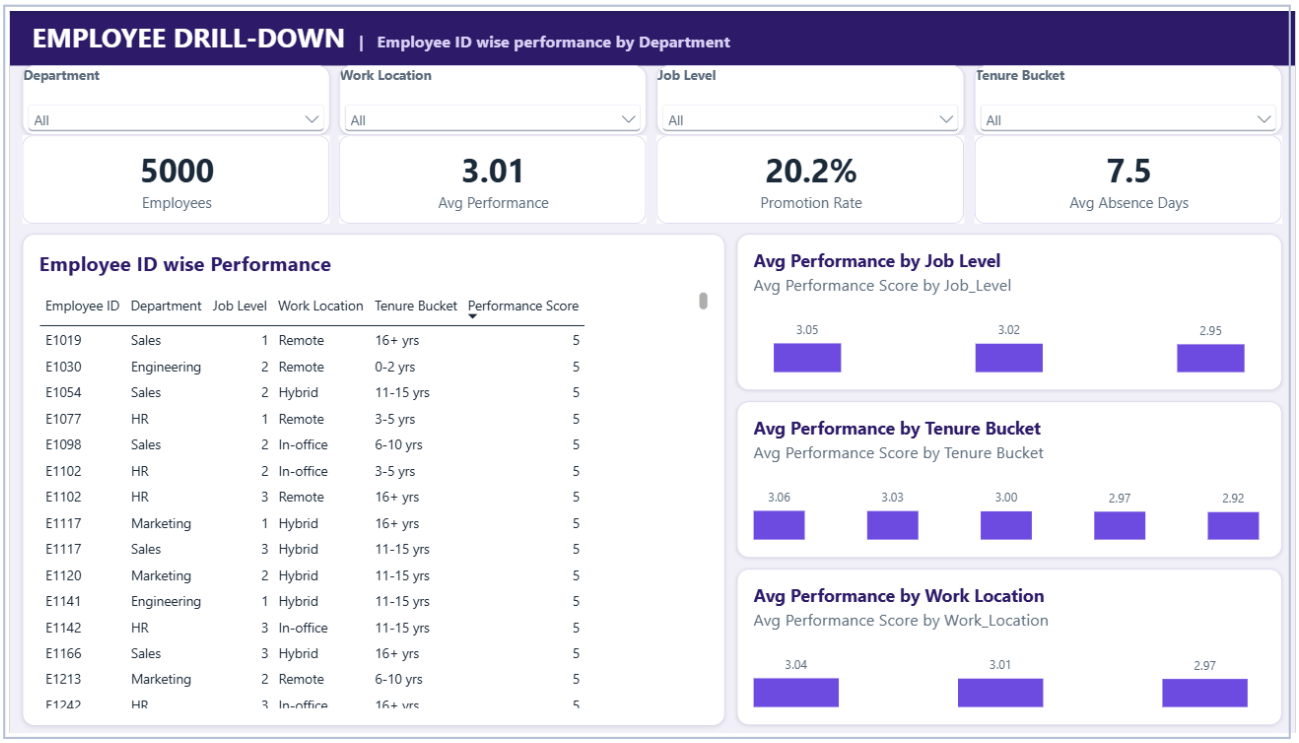


Figure 4: Employee Drill-down page

7. Key Findings

Key figures:

Total Employees	Avg Performance	Avg Salary	Avg Absence Days	Promotion Rate
5,000	3.01	5,972	7.5 days	20.2%

Overview – Who works here?

- The workforce is evenly spread: HR 1,270, Marketing 1,258, Engineering 1,243 and Sales 1,229 employees.
- Gender is almost equally split: Non-binary 1,684 (33.7%), Female 1,669 (33.4%) and Male 1,647 (32.9%).
- Work location is balanced: Hybrid 1,696, In-office 1,673 and Remote 1,631.
- Job levels hold 1,665 (Level 1), 1,706 (Level 2) and 1,629 (Level 3) employees; performance scores 1 to 5 each hold about 1,000 employees.
- The workforce is experienced: 6-10 yrs (1,302), 16+ yrs (1,276) and 11-15 yrs (1,193) dominate, while only 474 employees have 0-2 years.

Performance – How well are teams performing?

- Overall average performance is 3.01. Department scores are Engineering 3.03, Sales 3.03, HR 3.02 and Marketing 2.95, so Marketing is the lowest rated.
- Average absence is 7.5 days. It varies little by department (about 7.37 to 7.51 days) and shows no clear straight-line link with the performance score (7.11 to 7.93 days).
- Average performance falls with job level (Level 1: 3.05, Level 2: 3.02, Level 3: 2.95) and with tenure (3.06 for 0-2 yrs down to 2.92 for 16+ yrs).
- *Note: the Performance page in Figure 2 was captured with the Gender slicer set to Non-binary, so its values are for that group (1,684 employees) rather than the whole company.*

Pay & Promotion

- Average salary is nearly flat across job levels (5,983 / 5,976 / 5,957), so higher levels are not paid noticeably more.
- Salary by performance score peaks at scores 2 and 3 (6,002 and 6,012) rather than at score 5 (5,987), so pay is not clearly linked to performance.
- There is no meaningful gender pay gap: Female 5,974, Male 5,972, Non-binary 5,972.
- Marketing has the highest average salary (6,008) and the highest promotion rate (22.18%), although it has the lowest performance score; Sales 19.77%, Engineering 19.63% and HR 19.29% follow.
- Promotion rate is highest for new joiners (21.31% for 0-2 yrs) and lowest for the longest-serving (19.36% for 16+ yrs).

Department	Total Employees	Avg Performance	Average Salary
Engineering	1,243	3.03	5,939.01
HR	1,270	3.02	5,944.69
Marketing	1,258	2.95	6,008.22
Sales	1,229	3.03	5,997.80
Total	**5,000**	**3.01**	**5,972.32**

Employee Drill-down

- The drill-down page lists every employee with department, job level, location, tenure bucket and performance score, so individual cases (for example, employees rated 5) can be located and compared against the group.
- Average performance by tenure bucket declines steadily: 3.06, 3.03, 3.00, 2.97 and 2.92 from the 0-2 yrs to the 16+ yrs band.

8. Conclusion

The Employee Performance Analytics Dashboard turns a 5,000-record employee table into four connected pages that answer the main workforce questions: who works here, how teams perform, how people are paid and promoted, and how an individual compares with the group. The analysis shows a very balanced organisation by department, gender and location, an average performance of 3.01 and a promotion rate of 20.2%. It also points to areas for attention: pay is only weakly linked to job level and performance, Marketing is rewarded most but rated lowest, and performance declines with tenure and seniority. These insights can help HR review the salary structure, tighten promotion criteria and plan engagement for long-serving staff. The project demonstrates data cleaning, data modelling, DAX measures and clear dashboard design in Power BI.